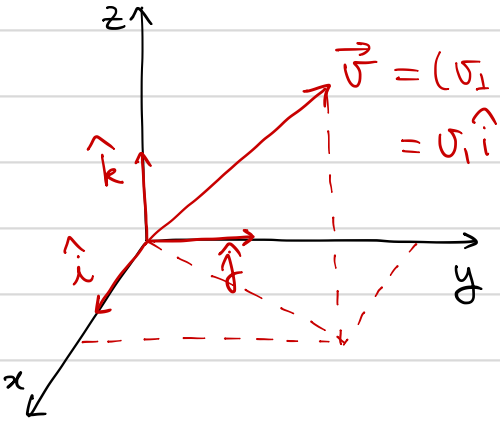
Lecture 4

- Review of vectors (12.3–4)
- Surface area (15.5)



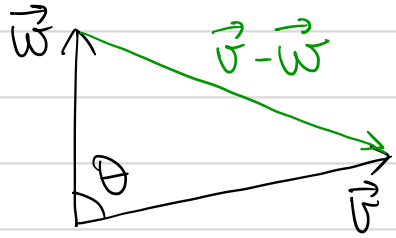
$$\vec{v} = (v_1, v_2, v_3) = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$$

$$\hat{i} = (1,0,0)$$
$$\hat{j} = (0,1,0)$$
$$\hat{k} = (0,0,1)$$

$$\text{length}^2 = |\vec{v}|^2 = \sqrt{v_1^2 + v_2^2}^{\,2} + v_3^2 = v_1^2 + v_2^2 + v_3^2 = \vec{v}\cdot\vec{v}$$

Dot product:

$$\vec{v}\cdot\vec{w} = v_1w_1 + v_2w_2 + v_3w_3 = |\vec{v}|\cdot|\vec{w}|\cdot\cos\theta$$



Why:

Law of cosines (see supplementary notes)

$$|\vec{v}-\vec{w}|^2 = |\vec{v}|^2 + |\vec{w}|^2 - 2|\vec{v}||\vec{w}|\cos\theta$$

$$\parallel$$

$$(\vec{v}-\vec{w})\cdot(\vec{v}-\vec{w}) = \vec{v}\cdot\vec{v} + \vec{w}\cdot\vec{w} - 2\vec{v}\cdot\vec{w}$$

Cross product:

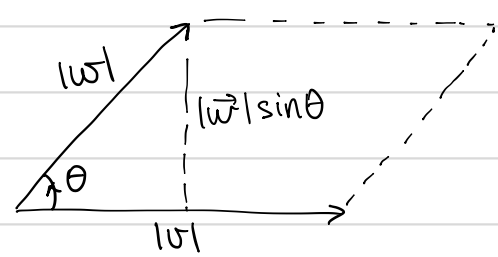
$$\vec{v}\times\vec{w} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} \begin{matrix} \hat{i} & \hat{j} \\ v_1 & v_2 \\ w_1 & w_2 \end{matrix}$$

$$= (v_2w_3 - v_3w_2)\hat{i} + (v_3w_1 - v_1w_3)\hat{j} + (v_1w_2 - v_2w_1)\hat{k}$$

Facts:

① $|\vec{v}\times\vec{w}| = |\vec{v}||\vec{w}|\sin\theta$

= Area of paralelogram



Why:

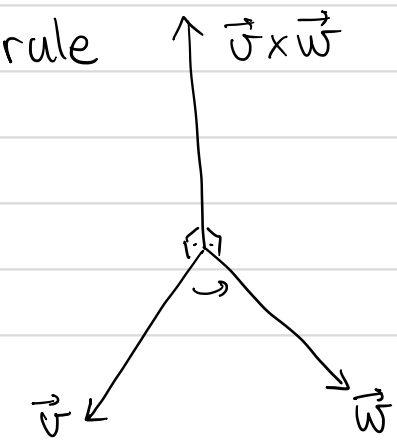
boring computation

$$|\vec{v}\times\vec{w}|^2 = |\vec{v}|^2|\vec{w}|^2 - (\vec{v}\cdot\vec{w})^2 = |\vec{v}|^2|\vec{w}|^2(1-\cos^2\theta) = |\vec{v}|^2|\vec{w}|^2\sin^2\theta \quad //$$

② Right hand rule

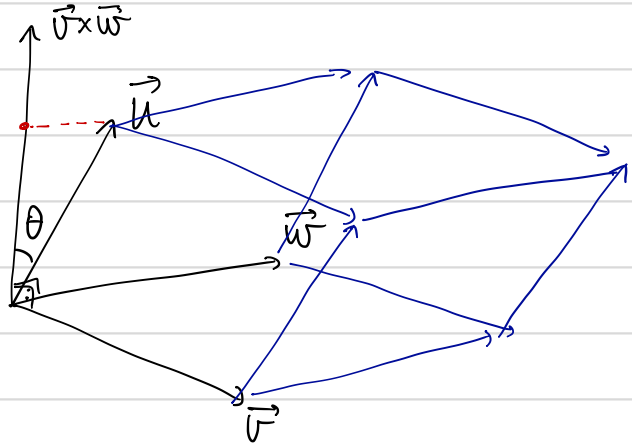
$$\vec{v}\cdot(\vec{v}\times\vec{w}) = 0$$
$$\vec{w}\cdot(\vec{v}\times\vec{w}) = 0$$

easy calculation



Determinant = ± Volume of box.

3D



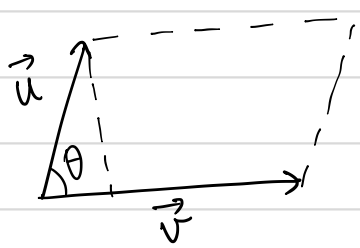
$$\begin{pmatrix}\text{Volume}\\ \text{of box}\end{pmatrix} = \pm(\text{Area parallelogram}) \cdot |\vec{u}|\cos\theta$$

$$= \pm\ \vec{u}\cdot(\vec{v}\times\vec{w})$$

$$= \pm \det\begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}$$

Sign is minus if $\vec{u}$ and $\vec{v}\times\vec{w}$ are in opposite sides of the plane spanned by $\vec{v}$ and $\vec{w}$.

2D

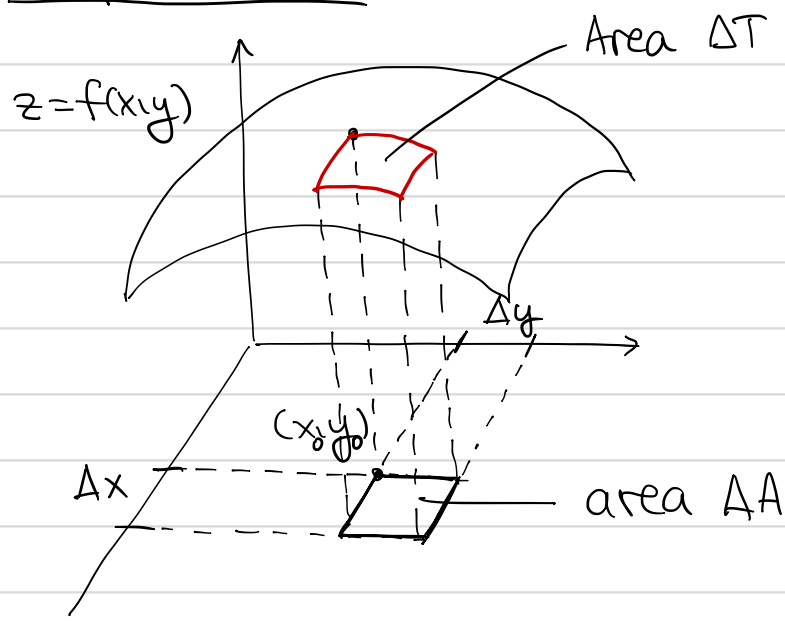


$$\text{Area} = \pm \det\begin{bmatrix} u_1 & u_2 \\ v_1 & v_2 \end{bmatrix}$$

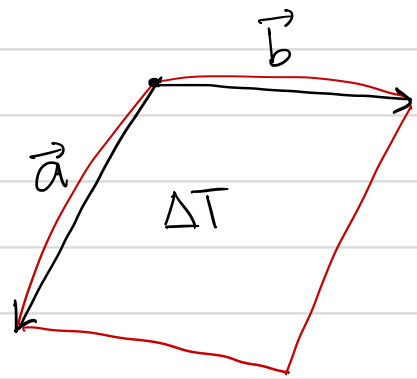
Why: $\vec{u},\vec{v}$ in the xy-plane

$\Rightarrow\ \vec{u}\times\vec{v} = \det\begin{pmatrix} u_1 & u_2 \\ v_1 & v_2 \end{pmatrix}\hat{k}$

Surface area



Zoom



$$\vec{a} = (\Delta x, 0, f(x_0+\Delta x, y_0) - f(x_0,y_0))$$
$$\approx \left(\Delta x, 0, \frac{\partial f}{\partial x}(x_0,y_0)\,\Delta x\right)$$

$$\vec{b} = (0, \Delta y, f(x_0, y_0+\Delta y) - f(x_0,y_0))$$
$$\approx \left(0, \Delta y, \frac{\partial f}{\partial y}(x_0,y_0)\,\Delta y\right)$$

$$\vec{a}\times\vec{b} \approx \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \Delta x & 0 & f_x \Delta x \\ 0 & \Delta y & f_y \Delta y \end{vmatrix}$$

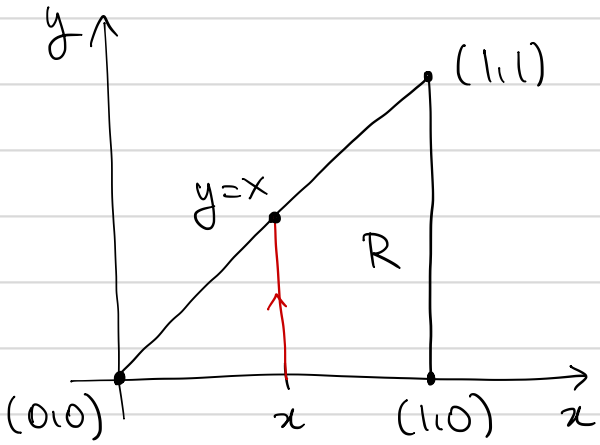
$$= (-f_x \Delta x \Delta y, -f_y \Delta x \Delta y, \Delta x \Delta y)$$

$$\Delta T \approx |\vec{a}\times\vec{b}|$$
$$\approx |(-f_x, -f_y, 1)| \Delta x \Delta y$$
$$= \sqrt{f_x^2 + f_y^2 + 1}\, \Delta x \Delta y$$

Area of $z = f(x,y)$ over region R of xy-plane:

$$\boxed{\text{Surface Area} = \iint_R \sqrt{f_x^2 + f_y^2 + 1}\, dA}$$

<u>Example</u>: $f(x,y) = x^2 + 2y$



$$A = \iint_R \sqrt{(2x)^2 + (2)^2 + 1}\, dA$$

$$= \int_0^1 \int_0^x \sqrt{4x^2+5}\, dy\, dx$$

$$= \int_0^1 \sqrt{4x^2+5}\, x\, dx$$

$$= \frac{2}{3}\cdot\frac{1}{8}(4x^2+5)^{3/2}\Big|_0^1 = \frac{1}{12}(27 - 5\sqrt{5})$$